



RAJALAKSHMI INSTITUTE OF TECHNOLOGY
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MINI PROJECT REPORT

REGISTER NUMBER	2117240070169
NAME	LOHITH.R
PROJECT TITLE	Parking Slot Finder Application Using BFS Algorithm
DATE OF SUBMISSION	
FACULTY IN-CHARGE	Ms. Phebe Persis

Signature of Faculty In-charge

1. INTRODUCTION:

- In today's rapidly urbanizing world, finding an available parking space has become a significant challenge, especially in densely populated cities. The increasing number of vehicles and limited parking infrastructure often lead to congestion, time wastage, and driver frustration. To address this problem, intelligent parking management systems have gained attention as a practical solution to enhance parking efficiency and improve the overall driving experience.
 - This project, titled "**Parking Slot Finder Application using Bayes Theorem**", aims to develop an intelligent system that predicts the availability of parking slots based on probabilistic reasoning. By applying principles of **Bayesian inference**, the system calculates the likelihood of a parking slot being available at a given time and location, based on historical data and real-time observations.
 - Bayes Theorem provides a mathematical framework for updating probabilities as new information becomes available. In this context, it helps the application refine its predictions about parking slot availability by continuously learning from user inputs, time patterns, and sensor data. This probabilistic approach ensures that predictions are not just static but dynamically adjusted to real-world conditions.
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2. PROBLEM STATEMENT:

This project models a parking lot as a 2D grid, where each cell represents a parking space. The AI agent must navigate from the **entry point 'I'** to the **nearest empty slot 'E'** using BFS.

Expected Results:

- BFS identifies the shortest and most efficient path to an empty slot.
- The system marks the slot as occupied once parked.

Future Possibilities:

- Add multiple entry points and dynamic slot updates.
- Extend with A* Search or integrate a graphical interface using Tkinter or Pygame.
- Include real-time sensors and IoT integration for live parking data.

3. THEORETICAL BACKGROUND:

Depth-First Search (DFS):

- Explores deep paths first.
- May not find the shortest path in a grid.

Breadth-First Search (BFS):

- Explores level by level.
- Guarantees the shortest path in unweighted grids.
- Designed for weighted graphs.
- Dijkstra/ A* search
- Not required for this project since all moves are equal.

Justification for Choosing BFS:

- The parking lot is modeled as an **unweighted grid**.
- BFS ensures the **shortest path** to the nearest empty slot.
- DFS may explore unnecessary paths or miss optimal solutions.
- Weighted algorithms are unnecessary for equal-cost movements.

4. ALGORITHM EXPLANATION:

Breadth-First Search (BFS) is a grid-based traversal algorithm that explores cells in layers.

Steps:

1. Start at the entry point 'I'.
 2. Add it to a queue and mark it as visited.
 3. While the queue is not empty:
 - Remove the first cell.
 - If it is an empty slot 'E', stop and mark it as parked 'X'.
 - Otherwise, add all unvisited neighboring cells to the queue.
 4. Repeat until an empty slot is found or all reachable cells are explored.
-

5. BFS EXAMPLE GRID:

Initial Grid:

I X X E

X X E X

X E X X

X X X E

Execution Steps:

- Queue: [(0, 0)] → Explore right and down
- Queue: [(0, 3)] → Found empty slot
- Mark (0, 3) as parked 'X'

Shortest Path Found:

Start → (0, 3) → Parked

Updated Grid:

I X X X

X X E X

X E X X

X X X E

6. QUEUE ACCESS ORDER IN BFS:

In the Parking Slot Finder Application, the **Breadth-First Search (BFS)** algorithm uses a **queue** to explore the parking lot grid level by level. The queue ensures that the nearest positions are checked first, guaranteeing the shortest path to an empty slot.

Step-by-Step Access Order:

1. Initialize the Queue

- Add the starting position 'I' to the queue.
- Example: Queue = [(0, 0)]

2. Explore the First Position

- Remove the first position from the queue using popleft().
- Check if it is an empty slot 'E'.
- If yes, mark it as parked 'X' and stop.

3. Add Valid Neighbors

- If the current cell is not 'E', check its neighbors:
 - Up (r-1, c)
 - Down (r+1, c)
 - Left (r, c-1)
 - Right (r, c+1)
- For each neighbor:
 - If it is not blocked 'X' and not visited, add it to the queue.
 - Example: Queue = [(0, 3), (1, 0)]

4. Repeat Until Found

- Continue removing and exploring positions from the queue.
- BFS always explores the **oldest added position first**, ensuring the shortest path.

5. Mark the Slot

- Once an empty slot 'E' is found, change it to 'X' to show it is now parked.

IMPLEMENTATION AND CODE:

```
from collections import deque
```

```
# Step 1: Create a 10x10 parking lot grid with open paths
```

```
def create_parking_lot():
```

```
    lot = [  
  
        ['I', 'O', 'O', 'E', 'X', 'X', 'X', 'X', 'X', 'X'],  
  
        ['X', 'O', 'X', 'O', 'O', 'X', 'X', 'X', 'X', 'X'],
```

```

['X', 'O', 'X', 'X', 'O', 'O', 'O', 'X', 'X', 'X'],
['X', 'O', 'O', 'O', 'E', 'X', 'O', 'X', 'X', 'X'],
['X', 'X', 'X', 'X', 'O', 'O', 'O', 'X', 'X', 'X'],
['X', 'X', 'X', 'X', 'X', 'X', 'O', 'O', 'O', 'E'],
['X', 'X', 'X', 'X', 'X', 'X', 'X', 'X', 'O', 'X'],
['X', 'X', 'X', 'X', 'X', 'X', 'X', 'X', 'O', 'X'],
['X', 'X', 'X', 'X', 'X', 'X', 'X', 'X', 'O', 'X'],
['X', 'X', 'X', 'X', 'X', 'X', 'X', 'X', 'O', 'E']
]

return lot, (0, 0)

```

Step 2: Print the parking lot grid

```
def print_grid(lot, title):
```

```
    print(f"\n P {title}:")
```

```
    for row in lot:
```

```
        print(" ".join(row))
```

```
    print()
```

Step 3: BFS to find and mark the nearest empty slot

```
def find_and_park(lot, start):
```

```
    rows, cols = len(lot), len(lot[0])
```

```

visited = [[False]*cols for _ in range(rows)]

queue = deque([start])

visited[start[0]][start[1]] = True

directions = [(-1,0), (1,0), (0,-1), (0,1)] # Up, Down, Left, Right

while queue:

    r, c = queue.popleft()

    # Check if current cell is an empty slot
    if lot[r][c] == 'E':

        lot[r][c] = 'X' # Mark as occupied

        print(f"✅ Nearest free parking slot found at: ({r}, {c})")

        print(f"🚗 Vehicle parked successfully!")

        return True

    # Explore neighbors
    for dr, dc in directions:

        nr, nc = r + dr, c + dc

        if 0 <= nr < rows and 0 <= nc < cols:

            if not visited[nr][nc] and lot[nr][nc] in ('O', 'E'):

                visited[nr][nc] = True

```

```
queue.append((nr, nc))
```

```
print("❌ No empty parking slot available.")
```

```
return False
```

```
# Step 4: Remove (unpark) a vehicle
```

```
def remove_car(lot):
```

```
    print_grid(lot, "Current Parking Lot")
```

```
    try:
```

```
        r = int(input("Enter the row number of the car to remove (0–9): "))
```

```
        c = int(input("Enter the column number of the car to remove (0–9): "))
```

```
        if lot[r][c] == 'X':
```

```
            lot[r][c] = 'E'
```

```
            print(f"🚗 Car removed successfully from location ({r}, {c}).")
```

```
        else:
```

```
            print("❌ That spot doesn't have a car parked.")
```

```
    except (ValueError, IndexError):
```

```
        print("⚠️ Invalid input! Please enter row and column between 0–9.")
```

```
# Step 5: Menu system
```

```
def main_menu():  
    lot, start_pos = create_parking_lot()  
  
    while True:  
        print("\n=== 🚗 PARKING SLOT FINDER ===")  
        print("1. Find Nearest Parking Slot")  
        print("2. Remove a Parked Car")  
        print("3. Exit")  
        choice = input("Enter your choice (1-3): ").strip()  
  
        if choice == '1':  
            print_grid(lot, "Current Parking Lot")  
            find_and_park(lot, start_pos)  
            print_grid(lot, "Updated Parking Lot After Parking")  
        elif choice == '2':  
            remove_car(lot)  
            print_grid(lot, "Updated Parking Lot After Car Removal")  
        elif choice == '3':  
            print("👋 Exiting the program. Have a nice day!")  
            break  
        else:
```

```
print("❌ Invalid choice. Please enter 1, 2, or 3.")
```

```
# Run the program
```

```
if __name__ == "__main__":
```

```
    main_menu()
```

OUTPUT:

```
=== 🚗 PARKING SLOT FINDER ===  
1. Find Nearest Parking Slot  
2. Remove a Parked Car  
3. Exit
```

```
=== 🚗 PARKING SLOT FINDER ===  
1. Find Nearest Parking Slot  
2. Remove a Parked Car  
3. Exit  
Enter your choice (1-3): 1  
  
P Current Parking Lot:  
I O O E X X X X X  
X O X O O X X X X  
X O X X O O O X X  
X O O O E X O X X  
X X X X O O O X X  
X X X X X X O O E  
X X X X X X X O X  
X X X X X X X O X  
X X X X X X X O X  
X X X X X X X O E  
  
✅ Nearest free parking slot found at: (0, 3)  
🚗 Vehicle parked successfully!  
  
P Updated Parking Lot After Parking:  
I O O X X X X X X  
X O X O O X X X X  
X O X X O O O X X  
X O O O E X O X X  
X X X X O O O X X  
X X X X X X O O E  
X X X X X X X O X  
X X X X X X X O X  
X X X X X X X O X  
X X X X X X X O E
```

```
=== 🚗 PARKING SLOT FINDER ===
```

1. Find Nearest Parking Slot
2. Remove a Parked Car
3. Exit

Enter your choice (1-3): 2

PCurrent Parking Lot:

```
I O O X X X X X X
X O X O O X X X X
X O X X O O O X X
X O O O E X O X X
X X X X O O O X X
X X X X X O O O E
X X X X X X X O X
X X X X X X X O X
X X X X X X X O X
X X X X X X X O E
```

Enter the row number of the car to remove (0-9): 0

Enter the column number of the car to remove (0-9): 4

🚗 Car removed successfully from location (0, 4).

PUpdated Parking Lot After Car Removal:

```
I O O X E X X X X
X O X O O X X X X
X O X X O O O X X
X O O O E X O X X
X X X X O O O X X
X X X X X O O O E
X X X X X X X O X
X X X X X X X O X
X X X X X X X O X
X X X X X X X O E
```

```
=== 🚗 PARKING SLOT FINDER ===
```

1. Find Nearest Parking Slot
2. Remove a Parked Car
3. Exit

Enter your choice (1-3): 3

👋 Exiting the program. Have a nice day!

PS C:\Users\harsh\idlerc\New folder> █

Explanation:

- The project simulates a **smart parking management system** that helps find the **nearest available parking slot** from an entrance point using the **Breadth-First Search (BFS)** algorithm.
 - The parking area is represented as a **10×10 grid**, where:
 - 'I' denotes the **entry point**,
 - 'E' denotes an **empty parking slot**,
 - 'X' denotes an **occupied or blocked area**, and
 - 'O' denotes an **open path**.
 - The BFS algorithm explores all reachable cells **level by level** from the entry point to locate the **nearest free slot**.
 - Once found, the slot is marked as **occupied**, simulating that a vehicle has been parked successfully.
 - A **car removal feature** allows users to free an occupied slot and make it available again.
 - The system provides a **menu-based interface**, enabling users to:
 1. Find the nearest parking slot.
 2. Remove a parked car.
 3. Exit the program.
 - BFS ensures an **efficient and shortest route** to available slots while avoiding blocked cells.
-

RESULTS AND FUTURE ENHANCEMENTS:

Result

- BFS successfully identifies the **nearest available parking slot** in the grid.
- The algorithm handles **multiple branches** and ensures that no cell is revisited.
- The **menu-driven interface** allows users to park and unpark vehicles easily.
- The system simulates **real-time parking management**, showing grid updates after every operation.

- It demonstrates the efficiency and real-world applicability of BFS in **resource allocation and navigation systems**.

Future Enhancements

1. Multiple Vehicle Management

- Extend the system to manage **multiple vehicles simultaneously**, each tracked with unique IDs.
- Add a **queue system** to handle parking requests when the lot is full.

2. Smart Slot Reservation

- Introduce a **reservation feature** that allows users to pre-book parking slots.
- Use **priority queues** to assign slots based on vehicle type or user status.

3. Obstacles and Dynamic Movement

- Add **temporary obstacles** like maintenance zones or reserved spaces.
- Modify BFS to dynamically recalculate paths if obstacles appear.

4. Weighted Pathfinding

- Assign weights to certain cells (like narrow lanes or high-traffic areas).
- Upgrade the algorithm from BFS to **Dijkstra's** or **A*** to find optimal paths based on weights.

5. Real-Time Simulation and GUI

- Implement a **graphical interface** using **Tkinter** or **Pygame**.
- Visually display the parking grid and animate the car's movement to the allocated slot.

6. Database Integration

- Connect to a **MySQL or SQLite database** to store and retrieve slot information persistently.
- Keep track of total slots, parked vehicles, and available spaces across sessions.

GitHub Link:

 <https://github.com/Lohith156/Parking-slot-finder.git>

REFERENCES:

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